

The mechanism at every transported bank: self-duality of arbitrary representation data and the general Deligne chart

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Abstract

The transported banks of rung 7 — one bank per finite representation datum $\rho = \oplus_i \text{Sym}^{r_i}$ of the level-one seed, with compiled summability and L -factorization — receive their analytic packages. Two inputs make this uniform. First, *every* transported pair is self-dual: inverting every channel of a datum is the blockwise clock reversal, so the contragredient coefficients equal the primal coefficients at every ρ — one recursion, no per-datum argument. Second, the archimedean chart generalizes to the full Deligne form: an arbitrary finite list of real clocks convolved in signed-log height onto the complex-pair kernel, with Mellin multiplier exactly $\prod_j \Gamma_{\mathbb{R}}(s + \delta_j) \cdot \prod_i \Gamma_{\mathbb{C}}(s + \mu_i)$. The mechanism then holds at every datum and every chart type: one pointwise theta identity of the prescribed ρ -readouts yields entirety, the functional equation, and the chart identification for $L(\rho, s) = \prod_i L(\text{Sym}^{r_i}, s)$. Named instances land the remaining ladder banks at their classical charts: the degree-6 tensor $\text{std} \otimes \text{Sym}^2$ at datum $[3, 1]$, the degree-9 tensor $\text{Sym}^2 \otimes \text{Sym}^2$ at datum $[4, 2, 0]$, and split base change at datum $[1, 1]$ — each from *one* identity on its joint bank, with no per-factor input.

1 Self-duality of every transported datum

Rung 7’s transport (`UniformTransport.lean`) assigns to every finite list $\rho = [r_1, \dots, r_m]$ the channel type $\bigoplus_i \text{Fin}(r_i+1)$, the disjoint-union clock system, and the polynomial Satake pair `repSatakePair`, whose bank is the Dirichlet product $\prod_i c_{\text{Sym}^{r_i}}$ (`repBank.eq`). The new input is that this pair is its own contragredient at the coefficient level, for every datum:

Theorem 1 (`RepMechanism.repPair.dual.eq.primal`; machine-checked). *For every datum ρ and every j , the contragredient radial Euler coefficients of `repSatakePair` equal the primal ones. The proof is one recursion: the channel reversal $\sigma_\rho = \bigoplus_i \text{rev}_{\text{Fin}(r_i+1)}$ (`repPerm`) satisfies $w(x)^{-1} = w(\sigma_\rho x)$ on every channel (`repWeight.inv.eq`: $\alpha^{-(r-2k)} = \alpha^{r-2(r-k)}$ blockwise), and symmetric functions are permutation-invariant (`radialGlobalSatakeCoeff.comp.equiv`).*

Because the recursive channel type makes each step *definitionally* a direct sum, the induction consumes no reindexing: the case split is the case split of `Sum.elim`.

*Machine-checked companions: `RequestProject/GlobalHelixGeneralChartCompletion.lean` (the general kernel and its 3D bank layer, 26 named declarations) and `RequestProject/RepMechanism.lean` (datum self-duality, the general mechanism, and the named instances, 28 named declarations), each declaration at axiom footprint `{propxt, Classical.choice, Quot.sound}` or a subset, no `sorry`, no `axiom`, zero `sorryAx`.

2 The general Deligne chart

The complex-pair kernel ($\prod_i \Gamma_{\mathbb{C}}$) and the single-real mixed kernel ($\Gamma_{\mathbb{R}} \cdot \prod_i \Gamma_{\mathbb{C}}$) generalize by one list recursion (`GlobalHelixGeneralChartCompletion.lean`):

$$\text{generalKernelLog } [] \mu s = \text{completionKernelLog } \mu s,$$

$$\text{generalKernelLog } (\delta :: \delta s) \mu s = g_{\mathbb{R}}^{(\delta)} *_M \text{generalKernelLog } \delta s \mu s,$$

so the empty list is definitionally the complex kernel and a singleton the mixed kernel — both subsumptions are *compiled*, at the kernel and at the 3D banks (`generalKernelLog_nil`, `generalKernelLog_singleton`, and the corresponding readout identifications, the definitional ones by `rfl`). The three kernel facts hold at every real-shift list and every nonempty complex-shift list: uniform boundedness at every admissible weight, decay faster than every power, and the exact multiplier

$$\text{generalKernelLog_hasMellin} : \quad \tilde{g}(s) = \prod_j \Gamma_{\mathbb{R}}(s + \delta_j) \cdot \prod_i \Gamma_{\mathbb{C}}(s + \mu_i)$$

— the general Deligne factor at the archimedean place. The full 3D bank layer (primal and reciprocal-height contragredient readouts, theta identifications, local integrability, rapid decay, and both initial-half-plane identifications) is compiled on this kernel, uniformly in the two shift lists.

3 The mechanism at every datum and every chart

Theorem 2 (`RepMechanism.mechanismG`; machine-checked). *For every self-dual-multiset bank, every real-shift list with nonnegative real parts, every nonempty complex-shift list with nonnegative real parts, and every $\varepsilon \neq 0$: one pointwise theta identity between the prescribed general readouts yields the general package (`GeneralNicePackage`) — entire Λ , $\Lambda(1-s) = \varepsilon \Lambda(s)$, and $\Lambda(s) = \prod_j \Gamma_{\mathbb{R}}(s + \delta_j) \prod_i \Gamma_{\mathbb{C}}(s + \mu_i) \cdot L(s)$ on $\text{Re } s > B$ — with uniqueness of the continuation (`GeneralNicePackage.unique`), specialization to the complex and mixed packages at empty and singleton real-shift lists (`toNice`, `toMixed`), and the product law (`GeneralNicePackage.mul`: product bank, concatenated real and complex shift lists, product root number).*

Corollary 3 (the datum corollaries; machine-checked). *`rep_package_of_theta`,
`rep_mixed_package_of_theta`,
`rep_general_package_of_theta`*

For every representation datum ρ of the seed and every admissible chart — complex, mixed, or general — one theta identity of the prescribed ρ -readouts yields the complete package for the compiled product bank $L(\rho, s) = \prod_i L(\text{Sym}^{r_i}, s)$, at the uniform threshold of the datum. The self-duality input is Theorem 1; nothing is assumed per datum.

4 The remaining banks, at their classical charts

Three named instances land the ladder’s remaining banks. In each, the entire analytic superstructure of the *joint* object comes from one identity on the joint bank — no per-factor package, no seed input:

Corollary 4 (`tensor23_package_of_theta`; machine-checked). *Datum* $[3, 1]$ — the degree-6 tensor $\text{std} \otimes \text{Sym}^2$, whose bank is $c_{\text{Sym}^3} \star c_{\text{std}}$ (the Clebsch–Gordan factorization compiled at rung 4). *Chart*: the pure-complex $\Gamma_{\mathbb{C}}(s + \frac{k-1}{2})\Gamma_{\mathbb{C}}(s + \frac{3(k-1)}{2})\Gamma_{\mathbb{C}}(s + \frac{k-1}{2})$ at weight $k = 2n$ — the Sym^3 pair and the seed shift. One identity yields the package. (The previous finish of rung 4 consumed one identity and the compiled seed rung; this instance needs only the identity.)

Corollary 5 (`tensor33_package_of_theta`; machine-checked). *Datum* $[4, 2, 0]$ — the degree-9 tensor $\text{Sym}^2 \otimes \text{Sym}^2$, bank $c_{\text{Sym}^4} \star c_{\text{Sym}^2} \star \zeta$ (rung 5’s decomposition; the rank-zero constituent is ζ by `symrBank_zero_eq_zeta`, compiled). *Chart*: the general form with complex shifts pinned to the exponent pairs $k-1, 2(k-1)$ (from Sym^4) and $k-1$ (from Sym^2), and the real-shift list parametric with nonnegative real parts — classically three real clocks, one per even constituent ($\text{Sym}^4, \text{Sym}^2, \text{Sym}^0$), with shifts fixed by the middle-line parities; whichever admissible $(\delta s, \varepsilon)$ satisfies the identity yields the package. This is the first finish of the degree-9 object as one object: rung 5 end-to-end from a single equation.

Corollary 6 (`bc_split_package_of_theta`; machine-checked). *Datum* $[1, 1]$, the split quadratic base-change bank $c_{\text{std}} \star c_{\text{std}}$ — rung 1’s split factorization

$$L(\text{BC } \pi) = L(\pi) L(\pi \otimes \chi) \quad (\chi \text{ split}),$$

at the doubled seed chart $\Gamma_{\mathbb{C}}(s + \frac{k-1}{2})^2$.

5 Register

Remark 7 (`register`). **Proved (machine-checked)**. Everything named above: datum self-duality at every ρ ; the general kernel’s bounds, decay, and multiplier; the general 3D bank layer; the general mechanism with uniqueness; the datum corollaries at all three chart types; the three named instances. All at axiom footprint $\{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$, no sorry.

Consumed from the corpus. Rung 7’s transport (channel recursion, bank, summability), the mechanism engine of the companion note, the real and complex clocks, and the theta/Mellin layer — all compiled.

Cited, not formalized. The classical discharges of the per-object identity: Gelbart–Jacquet [1] and Shimura [2] at Sym^2 , Kim–Shahidi [3] and Kim [4] at $\text{Sym}^3, \text{Sym}^4$ (hence at the tensor data $[3, 1], [4, 2, 0]$), Langlands [5] at base change. Named for scope, not consumed.

Not claimed. The truth of the theta identity for any datum beyond the seed is not claimed; it is the isolated input, one pointwise equation per bank. The real-shift values of the degree-9 chart are not asserted; the theorem is uniform in them. Inert base change and automorphic induction keep their landed coefficient identities (rungs 1–2) and are not datum instances: their twisted components are not symmetric powers of the seed.

Falsifier. A transported datum whose contragredient coefficients differ from its primal coefficients at any index would refute Theorem 1; a general kernel whose Mellin multiplier deviates from $\prod \Gamma_{\mathbb{R}} \cdot \prod \Gamma_{\mathbb{C}}$ at an admissible point, or a datum satisfying the identity whose L -readout fails the package, would refute the mechanism.

References

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